

MATA37H3 — Week 03

Integration Techniques I: Substitution & Integration by Parts

Recap and orientation

FTC (Week 2) reduced every definite integral to “find an antiderivative, then subtract endpoint values.” This unit is the first of two building a toolkit for actually *finding* antiderivatives when the integrand isn’t already a basic form. Here we cover the two most fundamental techniques: **substitution** (undoing the chain rule) and **integration by parts** (undoing the product rule). Next week’s session adds trigonometric-integral identities and partial fractions to the toolkit.

Why this matters

Nearly every integral on a MATA37/MAT137-style exam is designed to test whether you can correctly *identify* the technique, not just execute it. Substitution and integration by parts are the two techniques you will reach for constantly — inside trig integrals, inside partial-fraction results, inside improper-integral evaluations, and throughout the series unit when checking that a power series’ antiderivative was computed correctly.

KEY FACT Integration by Substitution (Undoing the Chain Rule) (MAT137 Unit 9)

If F is an antiderivative of f , then $\frac{d}{dx}[F(g(x))] = f(g(x))g'(x)$ by the chain rule, so

$$\int f(g(x))g'(x)dx = F(g(x)) + C.$$

In practice: pick $u = g(x)$ (the “inside function”), compute $du = g'(x)dx$, and rewrite the whole integral in terms of u only — if any leftover x ’s remain after substituting, the wrong u was chosen (or the integral needs a different technique entirely).

KEY FACT Substitution in a Definite Integral (MAT137 Unit 9)

When evaluating $\int_a^b f(g(x))g'(x)dx$ by substitution $u = g(x)$, the **limits of integration must also change**, to $u = g(a)$ and $u = g(b)$:

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

A common but invalid shortcut is to substitute $u = g(x)$ throughout the integrand while leaving the limits as the original x -values — this produces a numeric answer that is often coincidentally close but is not justified by the change-of-variables theorem, and the *evaluation* step (plugging in numbers) must use whichever variable’s limits you are currently using.

KEY FACT Integration by Parts (OpenStax §3.1, Theorem 3.1)

If $u = f(x)$ and $v = g(x)$ have continuous derivatives, then

$$\int u dv = uv - \int v du.$$

This comes directly from integrating the product rule $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$. Use it when the integrand is a **product** of two functions where one part becomes simpler when differentiated (choose that part as u) and the other part is easy to integrate (that part is dv). A common mnemonic for choosing u is **LIATE** (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — earlier in the list is generally the better choice of u).

KEY FACT Repeated / Tabular Integration by Parts (MAT137 Unit 9; Dummit)

When the “algebraic” factor is a polynomial of degree $n \geq 2$ (e.g. $\int x^3 e^x dx$), integration by parts must be applied **repeatedly**, each time differentiating the polynomial part further, until it reaches degree 0. The **tabular method** organizes this bookkeeping: make a column of successive derivatives of the polynomial factor (down to 0) alongside a column of successive antiderivatives of the other factor, then combine diagonally with alternating signs. When the pattern instead **cycles back to the original integral** (as with $\int e^{ax} \sin(bx) dx$, where two rounds of parts return a multiple of the same integral), solve for the original integral algebraically rather than trying to apply parts a third time.

Worked Examples

EXAMPLE 1

Basic substitution

Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

Let $u = \sqrt{x} = x^{1/2}$, so $du = \frac{1}{2\sqrt{x}} dx$, i.e. $\frac{dx}{\sqrt{x}} = 2 du$. Then

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int \sin u \cdot 2 du = -2 \cos u + C = -2 \cos \sqrt{x} + C. \quad \blacksquare$$

EXAMPLE 2

Substitution in a definite integral — done correctly

Evaluate $I = \int_0^2 \sqrt{x^3 + 1} x^2 dx$.

Let $u = x^3 + 1$, so $du = 3x^2 dx$, i.e. $x^2 dx = \frac{1}{3} du$. **Changing the limits:** when $x = 0$, $u = 1$; when $x = 2$, $u = 9$. So

$$I = \int_1^9 \sqrt{u} \cdot \frac{1}{3} du = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} \Big|_1^9 = \frac{2}{9} (9^{3/2} - 1^{3/2}) = \frac{2}{9} (27 - 1) = \frac{52}{9}.$$

Note that the limits were changed to u -values (1 and 9), not left as 0 and 2 — mixing the new integrand with the old limits would be evaluating the wrong quantity. $\blacksquare$

EXAMPLE 3

Integration by parts, single application

Evaluate $\int x e^{-2x} dx$.

Choose $u = x$ (gets simpler when differentiated), $dv = e^{-2x} dx$. Then $du = dx$, $v = -\frac{1}{2}e^{-2x}$.
 $\int x e^{-2x} dx = uv - \int v du = -\frac{x}{2}e^{-2x} - \int (-\frac{1}{2}e^{-2x}) dx = -\frac{x}{2}e^{-2x} + \frac{1}{2} \int e^{-2x} dx = -\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x} + C$. ■

EXAMPLE 4

Repeated integration by parts

Evaluate $\int x^2 \sin x dx$.

First round: $u = x^2$, $dv = \sin x dx \Rightarrow du = 2x dx$, $v = -\cos x$:

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx.$$

Second round on $\int 2x \cos x dx$: $u = 2x$, $dv = \cos x dx \Rightarrow du = 2 dx$, $v = \sin x$:

$$\int 2x \cos x dx = 2x \sin x - \int 2 \sin x dx = 2x \sin x + 2 \cos x + C.$$

Combining, $\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$. ■

EXAMPLE 5

Tabular integration by parts — three rounds in one table

Evaluate $\int x^3 e^x dx$ using the tabular method.

Differentiate x^3 down to 0 in one column, and antidifferentiate e^x (which reproduces itself) in a parallel column, alternating signs $+, -, +, -$ down the diagonal products:

Sign	D -column (derivatives of x^3)	I -column (antiderivatives of e^x)
+	x^3	e^x
−	$3x^2$	e^x
+	$6x$	e^x
−	6	e^x
	0	e^x

Multiply each D -entry by the *next* I -entry down, with the listed sign, and stop once the D -column reaches 0:

$$\int x^3 e^x dx = x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C.$$

This is exactly three applications of integration by parts done in one pass — the tabular method is a bookkeeping shortcut, not a new theorem, and it only works cleanly when one column terminates at 0 (a polynomial) while the other column is easy to keep antidifferentiating (as e^x , $\sin x$, $\cos x$ all are). ■

EXAMPLE 6

The “solve for the integral” trick — a column that never terminates

Evaluate $I = \int e^x \sin x dx$.

Round 1: $u = e^x$, $dv = \sin x dx \Rightarrow du = e^x dx$, $v = -\cos x$:

$$I = -e^x \cos x + \int e^x \cos x dx.$$

Round 2, on $\int e^x \cos x dx$: $u = e^x$, $dv = \cos x dx \Rightarrow du = e^x dx$, $v = \sin x$:

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx = e^x \sin x - I.$$

Substituting back: $I = -e^x \cos x + e^x \sin x - I$, so $2I = e^x(\sin x - \cos x)$, giving

$$I = \int e^x \sin x \, dx = \frac{e^x(\sin x - \cos x)}{2} + C.$$

Here neither column of a tabular setup would ever reach 0 (differentiating/antidifferentiating e^x and $\sin x/\cos x$ cycles forever) — this is the signal to stop applying parts and instead solve the resulting algebraic equation for I . ■

EXAMPLE 7

Logarithm as the u — nothing else to differentiate

Evaluate $\int \ln x \, dx$.

Treat this as $\int 1 \cdot \ln x \, dx$: choose $u = \ln x$ (it must be u , since there is no antiderivative shortcut for $dv = \ln x \, dx$), $dv = dx$. Then $du = \frac{1}{x} dx$, $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C. \quad \blacksquare$$

EXAMPLE 8

LIATE with two genuinely different factors

Evaluate $\int x \arctan x \, dx$.

By LIATE, Inverse-trig outranks Algebraic, so $u = \arctan x$, $dv = x \, dx \Rightarrow du = \frac{1}{1+x^2} dx$,

$$v = \frac{x^2}{2}:$$

$$\int x \arctan x \, dx = \frac{x^2}{2} \arctan x - \int \frac{x^2}{2(1+x^2)} \, dx.$$

Simplify the leftover integral by dividing out **before** reaching for partial fractions: $\frac{x^2}{1+x^2} =$

$$1 - \frac{1}{1+x^2}, \text{ so}$$

$$\int x \arctan x \, dx = \frac{x^2}{2} \arctan x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx = \frac{x^2}{2} \arctan x - \frac{x}{2} + \frac{1}{2} \arctan x + C. \quad \blacksquare$$

EXAMPLE 9

Level: Foundational — a trig antiderivative building on Practice Problem 2's cousin

Evaluate $\int \tan x \, dx$.

Rewrite $\tan x = \frac{\sin x}{\cos x}$ and let $u = \cos x$, so $du = -\sin x \, dx$, i.e. $\sin x \, dx = -du$. Then

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = \int \frac{-du}{u} = -\ln |u| + C = -\ln |\cos x| + C = \ln |\sec x| + C.$$

This is the “twin” of Practice Problem 2 ($\cot x$) — same substitution idea, opposite trig ratio — and the two answers are easy to mix up if the sign or the inside function gets copied on autopilot. ■

EXAMPLE 10

Level: Intermediate — substitution with algebraic rewriting first

Evaluate $\int x^3 \sqrt{x^2 + 1} dx$.

The obvious substitution $u = x^2 + 1$ gives $du = 2x dx$, which only accounts for **one** factor of x — but the integrand has $x^3 = x^2 \cdot x$. Solve for the leftover x^2 in terms of u : since $u = x^2 + 1$, $x^2 = u - 1$. Then

$$x^3 \sqrt{x^2 + 1} dx = x^2 \sqrt{x^2 + 1} \cdot x dx = (u - 1) \sqrt{u} \cdot \frac{du}{2}.$$

$\int x^3 \sqrt{x^2 + 1} dx = \frac{1}{2} \int (u^{3/2} - u^{1/2}) du = \frac{1}{2} \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) + C = \frac{1}{5} (x^2 + 1)^{5/2} - \frac{1}{3} (x^2 + 1)^{3/2} + C$.
The key move is substituting for the *leftover* power of x in terms of u , rather than declaring the integral “un-doable by substitution” the moment more than one factor of x remains after forming du . ■

EXAMPLE 11

Level: Advanced — integration by parts on a definite integral

Evaluate $\int_0^{\pi/2} x \cos x dx$.

Choose $u = x$, $dv = \cos x dx \Rightarrow du = dx$, $v = \sin x$:

$$\begin{aligned} \int_0^{\pi/2} x \cos x dx &= \left[x \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x dx = \left(\frac{\pi}{2} \cdot 1 - 0 \right) - \left[-\cos x \right]_0^{\pi/2} \\ &= \frac{\pi}{2} - (-\cos(\pi/2) + \cos 0) = \frac{\pi}{2} - (0 + 1) = \frac{\pi}{2} - 1. \end{aligned}$$

When integration by parts meets a definite integral, the boundary term $[uv]_a^b$ is evaluated at the endpoints **immediately**, and only the remaining integral $\int v du$ carries limits forward — there is no need to find a general antiderivative first and evaluate everything at the very end. ■

EXAMPLE 12

Level: Challenge — substitution feeds into tabular parts

Evaluate $\int x^5 e^{x^2} dx$.

Neither technique works alone: substitution’s obvious candidate $u = x^2$ leaves a leftover x^4 that doesn’t match any single du , and parts with x^5 as the polynomial factor would need five rounds against e^{x^2} — whose antiderivative isn’t even elementary, so parts cannot get started at all. Instead substitute on **part** of the integrand: let $u = x^2$, $du = 2x dx$, so $x^4 = u^2$ and $x dx = \frac{1}{2} du$:

$$\int x^5 e^{x^2} dx = \int x^4 e^{x^2} \cdot x dx = \int u^2 e^u \cdot \frac{du}{2} = \frac{1}{2} \int u^2 e^u du.$$

Now $\int u^2 e^u du$ is a textbook tabular-parts integral (compare Example 5): D -column $u^2, 2u, 2, 0$; I -column e^u, e^u, e^u, e^u ; signs $+, -, +$:

$$\begin{aligned} \int u^2 e^u du &= u^2 e^u - 2u e^u + 2e^u + C. \\ \int x^5 e^{x^2} dx &= \frac{1}{2} (x^4 e^{x^2} - 2x^2 e^{x^2} + 2e^{x^2}) + C = \frac{1}{2} e^{x^2} (x^4 - 2x^2 + 2) + C. \end{aligned}$$

The diagnostic skill here is recognizing that a *partial* substitution — one that simplifies the exponent while deliberately leaving a polynomial factor behind — can turn an integral that resists both techniques individually into one that yields cleanly to parts. ■

Practice Problems

Substitution

1. Evaluate $\int e^x \cos(e^x) dx$.

Your working:

2. Evaluate $\int \cot x dx$.

Your working:

3. Evaluate $\int_0^1 x e^{-x^2} dx$, carefully converting the limits of integration.

Your working:

4. The following write-up contains an error even though its final numeric answer happens to be wrong: " $\int_0^2 \sqrt{x^3 + 1} x^2 dx$: let $u = x^3 + 1$, $du = 3x^2 dx$; $= \frac{1}{3} \int_0^2 \sqrt{u} du = \left. \frac{2}{9} u^{3/2} \right|_0^2$." Identify the specific error and compute the correct value.

Your working:

5. Evaluate $\int \frac{x}{\sqrt{4 - x^2}} dx$.

Your working:

Integration by Parts — single and repeated

6. Evaluate $\int x e^{3x} dx$.

Your working:

7. Evaluate $\int x^2 \ln x \, dx$.

Your working:

8. Evaluate $\int x^4 e^{-x} \, dx$ using the tabular method from Example 5.

Your working:

9. Evaluate $\int e^{ax} \cos(bx) \, dx$ (constants $a, b \neq 0$), using the “solve for the integral” trick from Worked Example 6.

Your working:

Mixed / Applications

10. Estimate $\int_0^1 f'(x) dx$ and $\int_0^3 x f'(x) dx$ from a graph where f is a smooth increasing convex function with $f(0) = 2$, $f(1) = 3$, $f(3) \approx 4.7$ (using $\int_0^1 f'(x) dx = f(1) - f(0)$ via FTC Part 2, and integration by parts on the second one).

Your working:

11. Evaluate $\int \sin(\ln x) dx$.

Your working:

12. A student claims that $\int \frac{1}{x} \ln x dx$ “obviously needs integration by parts since it’s a product of $\frac{1}{x}$ and $\ln x$.” Show that a substitution is actually faster, and evaluate the integral both ways to confirm they agree.

Your working:

More Practice, Graded by Level and Idea

13. (*Level: Foundational — recognition*) Evaluate $\int \sec^2(3x) dx$.

Your working:

14. (*Level: Foundational — inverse trig via parts*) Evaluate $\int \arcsin x dx$.

Your working:

15. (*Level: Intermediate*) Evaluate $\int_1^e \ln x dx$ — the definite-integral version of Worked Example 7.

Your working:

16. (Level: Intermediate — algebra before technique) Evaluate $\int \frac{x^3}{x^2 + 1} dx$.

Your working:

17. (Level: Advanced) Evaluate $\int x \cos(2x) dx$ by parts, then verify your answer by differentiating it.

Your working:

18. (Level: Advanced — diagnose the error) A student evaluating $\int x e^{x^2} dx$ writes: “Let $u = x$, $dv = e^{x^2} dx$; then $v = e^{x^2}$, so $\int x e^{x^2} dx = x e^{x^2} - \int e^{x^2} dx$.” Explain precisely what is wrong with this approach — there are two separate problems with it — and give the correct evaluation.

Your working:

19. (*Level: Challenge — combined technique*) Evaluate $\int e^{\sqrt{x}} dx$.

Your working:

20. (*Level: Challenge — application*) Water flows through a pipe at a rate $r(t) = 5te^{-t/2}$ cubic meters per minute at time t minutes, for $t \geq 0$. Find the total volume, in cubic meters, that flows through the pipe during the first 4 minutes, i.e. compute $\int_0^4 5te^{-t/2} dt$.

Your working: